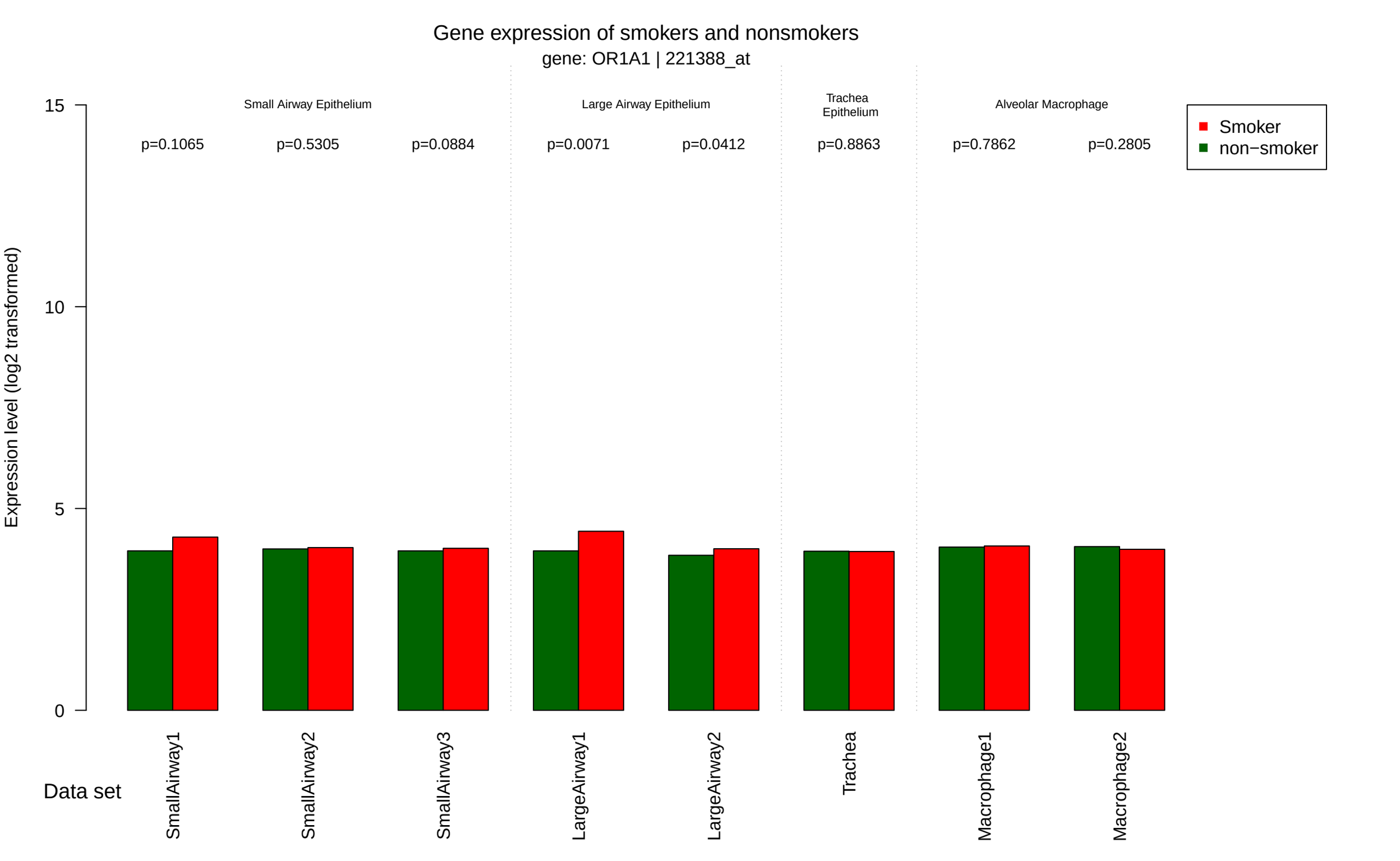
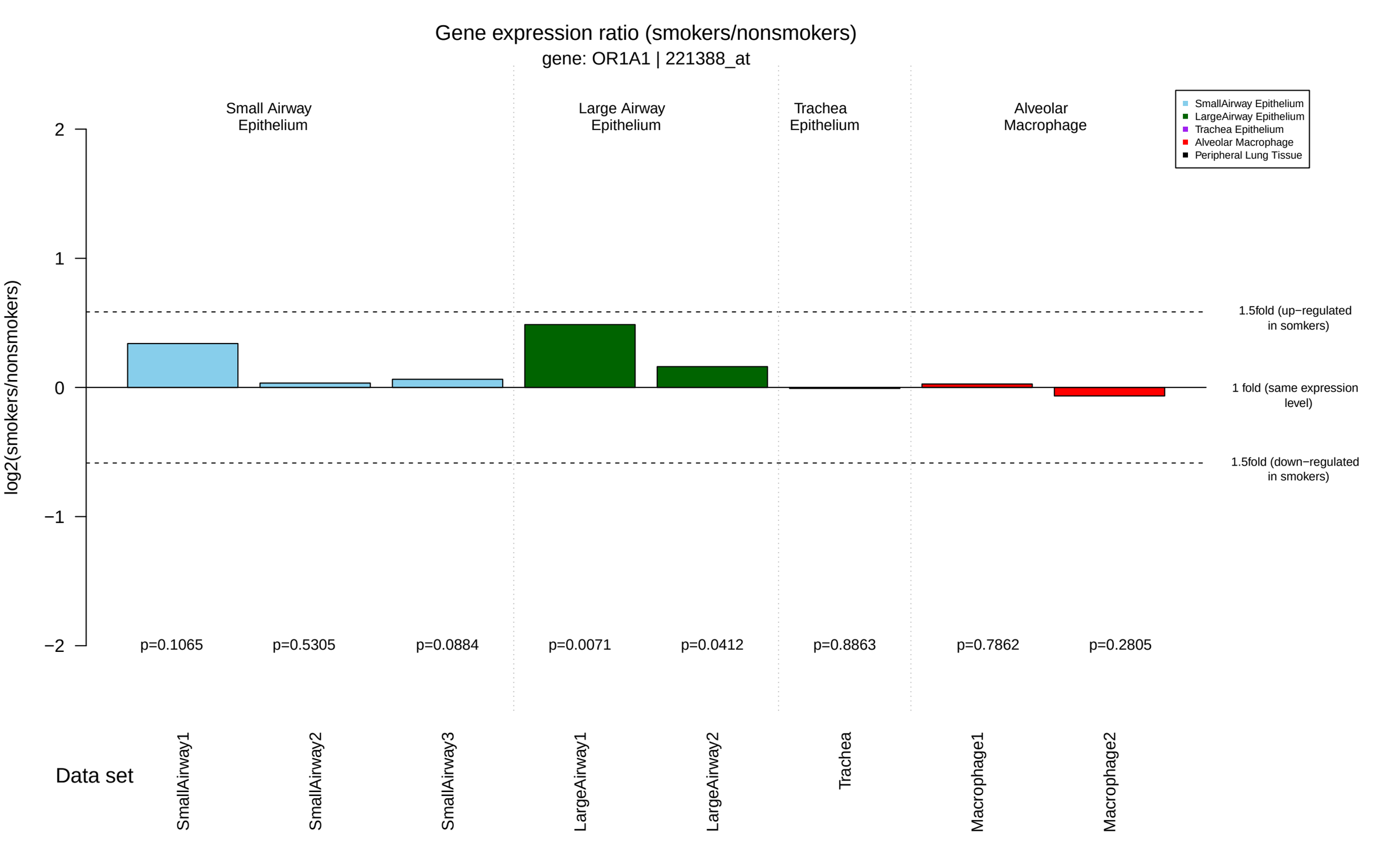
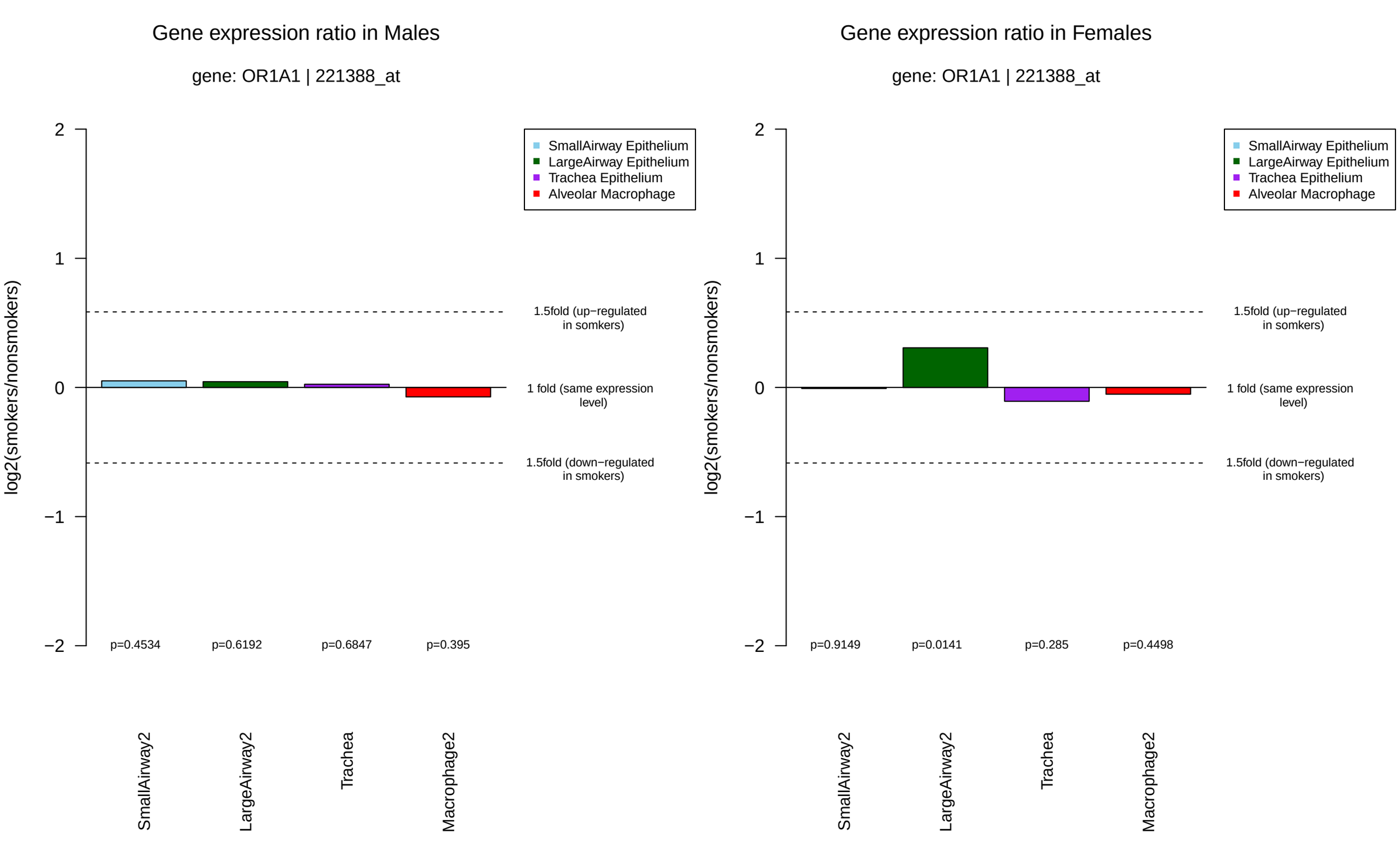
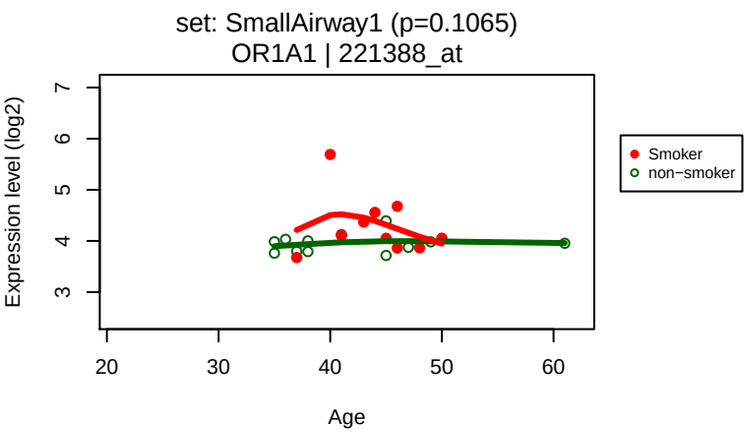
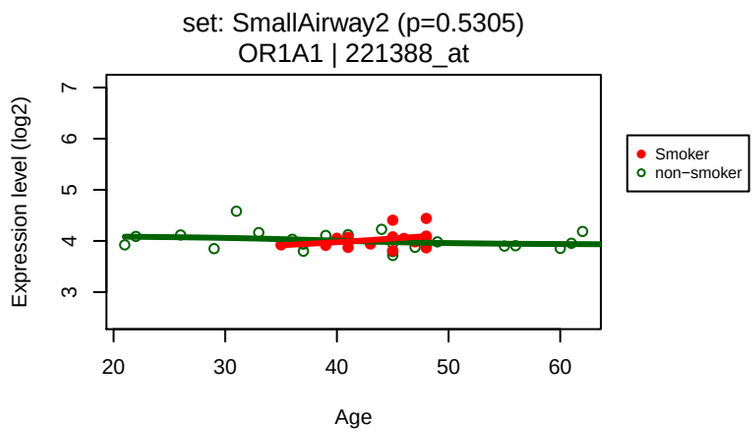
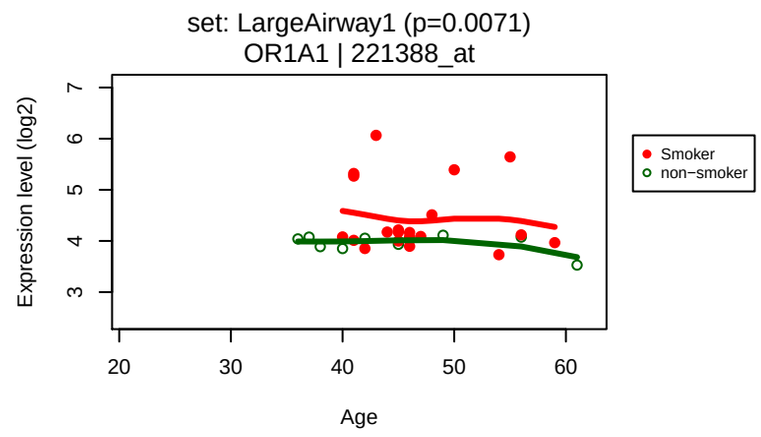

Gene expression of smokers and nonsmokers
gene: OR1A1 | 221388_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1065
p=0.5305
p=0.0884
p=0.0071
p=0.0412
p=0.8863
p=0.7862
p=0.2805
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

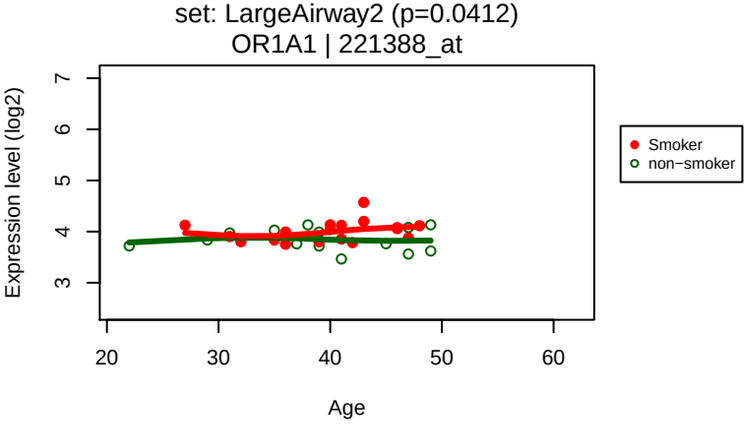

Gene expression ratio (smokers/nonsmokers)
gene: OR1A1 | 221388_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1065
p=0.5305
p=0.0884
p=0.0071
p=0.0412
p=0.8863
p=0.7862
p=0.2805
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

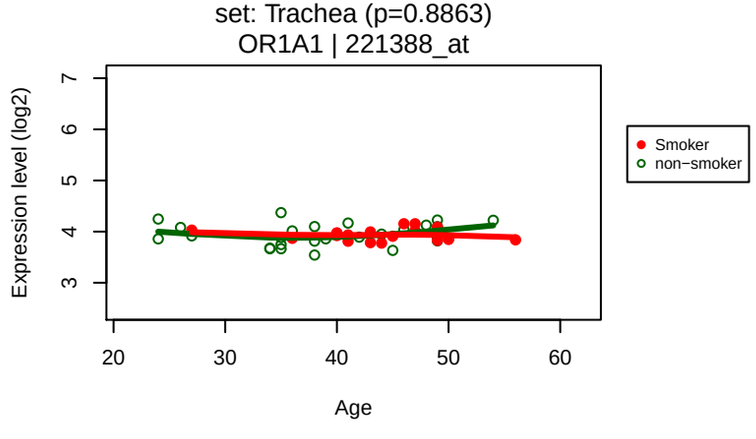

Gene expression ratio in Males
gene: OR1A1 | 221388_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4534
p=0.6192
p=0.6847
p=0.395
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR1A1 | 221388_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9149
p=0.0141
p=0.285
p=0.4498
SmallAirway2
LargeAirway2
Trachea
Macrophage2

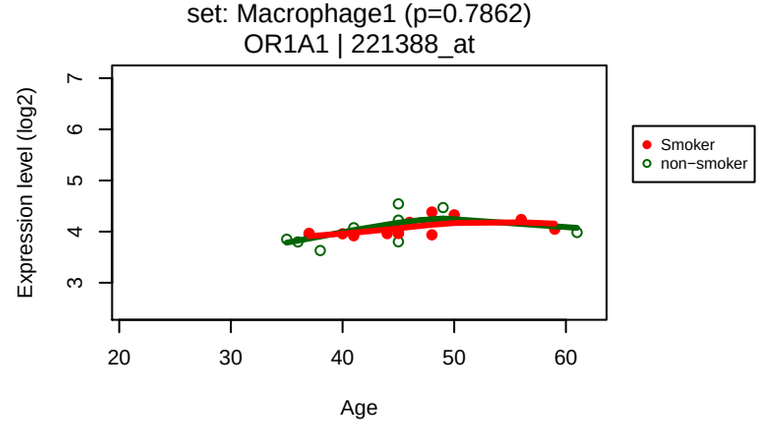

set: SmallAirway1 (p=0.1065)
OR1A1 | 221388_at
Expression level (log2)
Age
Smoker
non−smoker

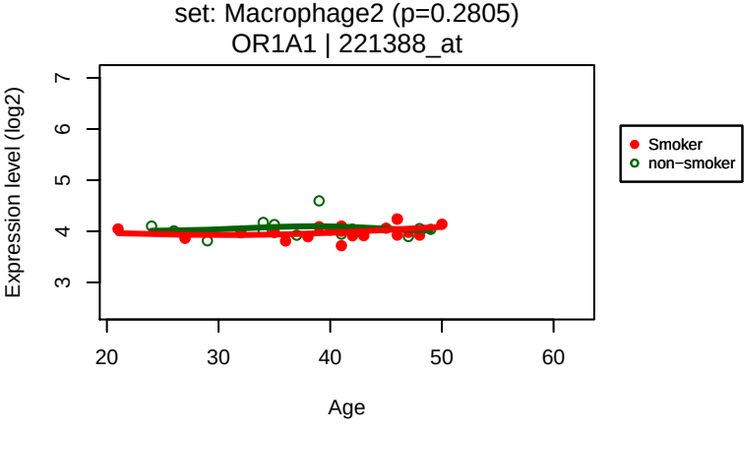

set: SmallAirway2 (p=0.5305)
OR1A1 | 221388_at
Expression level (log2)
Age
Smoker
non−smoker

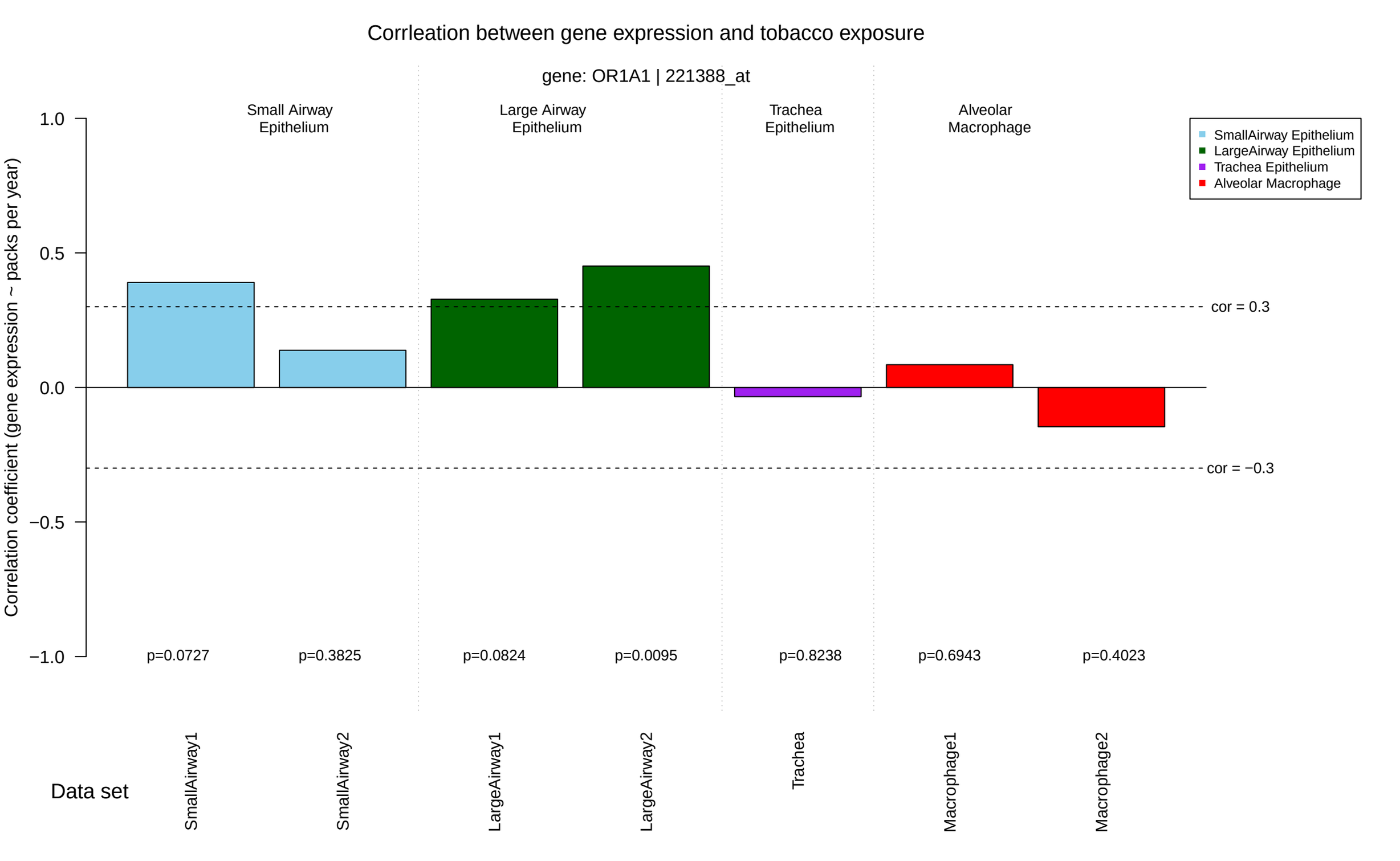

set: LargeAirway1 (p=0.0071)
OR1A1 | 221388_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0412)
OR1A1 | 221388_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8863)
OR1A1 | 221388_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7862)
OR1A1 | 221388_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2805)
OR1A1 | 221388_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR1A1 | 221388_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0727
p=0.3825
p=0.0824
p=0.0095
p=0.8238
p=0.6943
p=0.4023
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2